



Global asymptotic stability of limit cycle and $\mathcal{H}_2/\mathcal{H}_\infty$ performance of discrete-time switched affine systems[☆]

Lucas N. Egidio, Helder R. Daiha, Grace S. Deaecto^{*}

DMC - School of Mechanical Engineering, UNICAMP, 13083-860, Campinas, SP, Brazil

ARTICLE INFO

Article history:

Received 10 July 2019

Received in revised form 28 November 2019

Accepted 11 February 2020

Available online 19 March 2020

Keywords:

Switched affine systems

Discrete-time domain

Limit cycles

Global asymptotic stability

$\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance indexes

LMIs

ABSTRACT

This paper treats the control design of a state-dependent switching function for discrete-time switched affine systems, assuring global asymptotic stability of a limit cycle. This limit cycle is determined according to criteria of interest related to the steady-state behaviour of the system trajectories, as for instance, its mean value, oscillation amplitude and frequency. More specifically, based on a time-varying convex Lyapunov function, a min-type switching rule is determined in order to orchestrate the state trajectories, evolving from any initial condition, towards the desired limit cycle. Most of the available methodologies are able to guide the state trajectories to a set of attraction containing the equilibrium point, assuring its practical stability. However, although important, the behaviour of the trajectories inside this set is not considered. Differently, the proposed technique assures an appropriate steady-state behaviour by adequately designing the limit cycle. Moreover, the proposed conditions take into account $\mathcal{H}_2$ and $\mathcal{H}_\infty$ guaranteed costs, assuring also a suitable transient response. All the conditions are expressed in terms of linear matrix inequalities being simple to solve by readily available tools. Illustrative examples, one of them concerning the voltage regulation of a DC–DC multicellular converter, are used for validation and comparison.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

Switched affine systems are composed of a set of subsystems and a switching function (or switching rule) responsible for activating one of them at each instant of time. In our context this switching function is the control variable to be designed in order to assure stability and suitable performance. These systems are very common in several engineering areas, mainly in the power electronics domain, (Cardim, Teixeira, Assunção, & Covacic, 2009; Egidio, Daiha, Deaecto, & Geromel, 2017; Sferlazza, Albea-Sanchez, Martinez-Salamero, Garcia, & Alonso, 2020), and are characterised by the presence of affine terms, which are responsible for the existence of several equilibrium points. When the affine terms are null, the system becomes linear and the unique equilibrium point, common to all subsystems, is the

origin. For this simpler case, the literature presents several results dealing with the control design of a stabilising state or output-dependent switching function, see Allerhand and Shaked (2013), Daiha, Egidio, Deaecto, and Geromel (2017), Deaecto, Geromel, and Daafouz (2011), Fiacchini and Jungers (2014) and Geromel and Colaneri (2006), among others. However, when at least one affine term is non-null, the control problem is more challenging, since the chosen equilibrium point is generally not common to any subsystem. Consequently, asymptotic stability of this point is only obtained with an arbitrarily high switching frequency, which may be impossible to implement in practical applications due to physical limitations. Deaecto (2016), Deaecto and Santos (2015) and Seatzu, Corona, Giua, and Bemporad (2006) deal with global asymptotic stability of a desired equilibrium point for switched affine systems in the continuous-time domain, without considering any constraint on the switching rate. When some upper bound is imposed to the switching frequency, asymptotic stability is no longer possible and practical stability is assured. This is the general case of discrete-time switched affine systems, where the switching rate is naturally limited by the discrete nature of the system. Important results dealing with practical stability are presented in Hauroigne, Riedinger, and Jung (2011), Hetel and Fridman (2013), Sanchez, Garcia, Sabrina, Heemels, and Zaccarian (2019) and Sferlazza et al. (2020), in the continuous-time domain, and in Deaecto and Geromel (2017), Egidio and Deaecto

[☆] This research was supported by the "National Council for Scientific and Technological Development (CNPq), Brazil", under grant 303499/2018-4, by the "São Paulo Research Foundation (FAPESP), Brazil", under grant 2017/20343-0 and by the "Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil", Finance Code 001. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Hernan Haimovich under the direction of Editor Richard Middleton.

^{*} Corresponding author.

E-mail addresses: egidio@fem.unicamp.br (L.N. Egidio), daiha@fem.unicamp.br (H.R. Daiha), grace@fem.unicamp.br (G.S. Deaecto).

(2019) in the discrete-time domain. For continuous and discrete-time switched nonlinear systems with multiple equilibria, [Veer and Poulakakis \(2019\)](#) present interesting results on practical stability. In all these references, the trajectories are guided to a sufficiently small set of attraction containing the equilibrium point of interest. However, their behaviour inside this set cannot be directly controlled, which may affect the system performance. Therefore, neither $\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance indexes can be taken into account, since they are defined only for asymptotically stable systems, see [Deaecto and Santos \(2015\)](#). Very recently, and only for the continuous-time domain, some references have treated the asymptotic stabilisation of a limit cycle to control the steady-state behaviour of the system, see [Benmiloud, Benalia, Djemai, and Defoort \(2019\)](#) that deals with local stabilisation and [Patino, Riedinger, and Ruiz \(2010\)](#), which is based on a predictive control approach.

The present paper proposes a min-type state-dependent switching function to assure global asymptotic stability of a limit cycle of interest. Firstly, a set of possible limit cycles related to periodic switching sequences is determined, and a subset of candidates is chosen from specific criteria related to the desired steady-state behaviour of the system trajectories. Afterwards, a min-type switching function is designed based on a time-varying convex Lyapunov function in order to assure global asymptotic stability of a suitable limit cycle, as well as an $\mathcal{H}_2$ or $\mathcal{H}_\infty$ guaranteed cost. Compared to the existent techniques, our main contributions can be summarised as follows. First, to the best of the authors' knowledge there are only few results dealing with asymptotic stability of a limit cycle for switched affine systems, being the present paper, the first in the discrete-time domain. This allows us to control the steady-state behaviour of the trajectories, which is not possible with the methodologies dealing with practical stability. The second contribution is to take into account $\mathcal{H}_2$ and $\mathcal{H}_\infty$ guaranteed costs assuring a suitable performance of the transient response, which until now has been considered only for the continuous-time case, see [Deaecto \(2016\)](#), [Deaecto and Santos \(2015\)](#). The third one is related to conservativeness, which has been reduced by adopting a time-varying Lyapunov function. An illustrative example and a practical application in the power electronics domain are used for comparison and validation.

The notation used throughout is standard. For real vectors or matrices, $(\cdot)'$ refers to their transpose. For symmetric matrices, $(\bullet)$ denotes each of their symmetric blocks. The sets of real and natural numbers are $\mathbb{R}$ and $\mathbb{N}$, respectively. The set $\mathbb{K} = \{1, \dots, N\}$ is composed of the N first positive natural numbers. For any symmetric matrix, $X > (<) 0$ denotes a positive (negative) definite matrix. The trace of a matrix X is denoted as $\text{tr}(X)$. The unit simplex is defined as $\Lambda = \{\lambda \in \mathbb{R}^N : \lambda_i \geq 0, \sum_{i \in \mathbb{K}} \lambda_i = 1\}$. The convex combination of matrices $X_i, \forall i \in \mathbb{K}$, is $X_\lambda = \sum_{i \in \mathbb{K}} \lambda_i X_i, \lambda \in \Lambda$. The square L_2 -norm of $z[n], n \in \mathbb{N}$, is $\|z\|_2^2 = \sum_{n \in \mathbb{N}} \|z[n]\|^2$ where $\|z\|^2 = z'z$ is the square of the Euclidean norm. The set of all trajectories z such that $\|z\|_2^2 < \infty$ is denoted by L_2 . The ∞ -norm is defined as $\|z\|_\infty = \max_i |z_i|$. The modulo operator is defined as $c = a \bmod b$ where c is the remainder of the Euclidean division between the integers a and b . Finally, for a positive $\kappa \in \mathbb{N}$, the function $k(n) = n \bmod \kappa$ will be largely used throughout the paper to ease the notation.

2. Problem statement

Consider a discrete-time switched affine system given by

$$x[n+1] = A_{\sigma[n]}x[n] + b_{\sigma[n]} + H_{\sigma[n]}w[n] \quad (1)$$

$$z[n] = E_{\sigma[n]}x[n] + G_{\sigma[n]}w[n] \quad (2)$$

defined for all $n \in \mathbb{N}_- = \mathbb{N} \cup \{-1\}$, where $x[n] \in \mathbb{R}^{n_x}$ is the state, $w[n] \in \mathbb{R}^{n_w}$ is the exogenous input, $z[n] \in \mathbb{R}^{n_z}$ is the controlled

output and $\sigma[n] : \mathbb{N}_- \rightarrow \mathbb{K}$ is the switching sequence that selects one of the subsystems $\mathcal{G}_i = \{A_i, b_i, H_i, E_i, G_i\}, i \in \mathbb{K}$, as active at each instant of time.

Let us assume that system (1)–(2) with $w[n] = 0, \forall n \in \mathbb{N}_-$, admits a periodic solution $x_e[n]$ with period $\kappa > 0$ chosen by the designer, and that evolves from $x[-1] = x_e[-1]$. The trajectory $x_e[n]$ associated to a periodic switching sequence $\sigma[n] = c[k(n)], \forall n \in \mathbb{N}_-$, for some sequence $c = (c[0], \dots, c[\kappa-1]) \in \mathbb{K}^\kappa$, must satisfy

$$x_e[n+1] = A_{c[k(n)]}x_e[n] + b_{c[k(n)]} \quad (3)$$

whose correspondent limit cycle is given by

$$\mathcal{X}_e(c) = \{x_e[k(n)] : (3), n \in \mathbb{N}_-\} \quad (4)$$

Notice that κ points of $x_e[n]$ can be obtained from (3) for $n \in \{0, \dots, \kappa-1\}$ respecting the boundary condition $x_e[0] = x_e[\kappa]$. The fundamental period, given by the sequence $(x_e[0], \dots, x_e[\kappa-1])$, is determined from one of the N^κ possible periodic switching sequences $\sigma[n] = c[k(n)]$, according to criteria related to the desired steady-state behaviour of the system trajectories, as it will be clear in the next section.

Defining the auxiliary state variable $\xi[n] = x[n] - x_e[n], x_e[n] \in \mathcal{X}_e(c)$, we obtain the alternative system

$$\xi[n+1] = A_{\sigma[n]}\xi[n] + \ell_{\sigma[n]}[n] + H_{\sigma[n]}w[n] \quad (5)$$

$$z_e[n] = E_{\sigma[n]}\xi[n] + G_{\sigma[n]}w[n] \quad (6)$$

evolving from $\xi[-1] = 0$, with $\ell_i[n] = A_i x_e[n] - x_e[n+1] + b_i, \forall i \in \mathbb{K}$, and $z_e[n] = z[n] - E_{\sigma[n]}x_e[n], \forall n \in \mathbb{N}_-$. Notice that, whenever $\xi[n] = 0$ the limit cycle $x_e[n]$ is reached by the solution of system (1)–(2).

Our goal is to design a state-dependent switching function $\sigma[n] = u(x[n]) : \mathbb{R}^{n_x} \rightarrow \mathbb{K}$ to assure global asymptotic stability of the limit cycle $\mathcal{X}_e(c)$. In other words, we want to design $\sigma[n] = u(x[n])$ in order to guarantee that the origin $\xi = 0$ of (5)–(6) be a globally asymptotically stable equilibrium point, assuring a suitable upper bound for the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance indexes given in [Deaecto, Fioravanti, and Geromel \(2013\)](#) as follows:

$\mathcal{H}_2$ performance index: Considering that the system (5)–(6), defined for all $n \in \mathbb{N}_-$ with $\xi[-1] = 0$, is asymptotically stable, this index is given by

$$J_2(\sigma) = \sum_{r=1}^{n_w} \|z_{er}\|_2^2 + e_r' G_{\sigma[-1]}' G_{\sigma[-1]} e_r \quad (7)$$

where $z_{er}[n]$ is the controlled output correspondent to the impulsive input $w[n] = e_r \delta[n+1]$ with vector $e_r, r \in \{1, \dots, n_w\}$, forming the standard basis for $\mathbb{R}^{n_w}$ and $\delta[n]$ being the impulse function.

$\mathcal{H}_\infty$ performance index: Considering that the system (5)–(6), defined for $n \in \mathbb{N}$ with $\xi[0] = 0$, is asymptotically stable, this index is given by

$$J_\infty(\sigma) = \sup_{w \in \mathcal{L}_2 \setminus \{0\}} \frac{\|z_e\|_2^2}{\|w\|_2^2} \quad (8)$$

As it has been already mentioned in [Deaecto et al. \(2013\)](#), for switched linear systems, these indexes equal the $\mathcal{H}_2$ or $\mathcal{H}_\infty$ squared norm of the i th subsystem, whenever the switching rule is kept constant, that is, $\sigma[n] = i, \forall n \in \mathbb{N}_-$. Moreover, they require that the system be asymptotically stable and, therefore, cannot be evaluated by control techniques available in the literature that deal with practical stability, where the state trajectories evolve towards a set of attraction containing the equilibrium point, see [Deaecto and Geromel \(2017\)](#) and [Egidio and Deaecto \(2019\)](#). This occurs because, adopting these methodologies, the L_2 -norm $\|z_e\|_2^2$ does not converge.

In this paper, the control design conditions are based on the following time-varying convex Lyapunov function

$$v(\xi[n], n) = \xi[n]'P[n]\xi[n] \quad (9)$$

with $P[n] = P[k(n)] > 0$, $n \in \{0, \dots, \kappa - 1\}$, to be determined. This function certainly provides less conservative conditions than the time-invariant one recurrent in the literature, see [Deaecto and Geromel \(2017\)](#), [Hetel and Fridman \(2013\)](#) and [Sferlazza et al. \(2020\)](#).

3. Main results

This section presents our main results. Firstly, a methodology to generate a set of candidate limit cycles is provided. Afterwards, a state-dependent switching function assuring global asymptotic stability of a desired limit cycle is presented together with an associated guaranteed cost. Finally, the design conditions are generalised to cope with $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control.

3.1. Limit cycle generation

In this subsection, the goal is to determine the trajectories $x_e[n]$ associated to the periodic sequence $\sigma[n] = c[k(n)] \in \mathbb{K}$. To this end, let us recall some definitions borrowed from [Deaecto and Geromel \(2018\)](#). For a given positive $\kappa \in \mathbb{N}$ that defines the period of the limit cycle, let $\mathcal{C}(\kappa) = \mathbb{K}^\kappa$ be the set obtained from the Cartesian product of $\mathbb{K}$ by itself κ times. This set contains N^κ elements $c \in \mathcal{C}(\kappa)$, which are finite sequences $c = (c[0], \dots, c[\kappa - 1])$. Associated to each $c \in \mathcal{C}(\kappa)$ there exists $\mathcal{X}_e(c)$ given in (4), whose κ points of $x_e[n]$, $n \in \{0, \dots, \kappa - 1\}$, can be obtained from (3) with $x_e[0] = x_e[\kappa]$ as the solution of the linear equation

$$\tilde{A}(c)\tilde{x}_e = -\tilde{b}(c) \quad (10)$$

where $\tilde{x}_e = [x_e[0]' \ x_e[1]' \ \dots \ x_e[\kappa - 1]']'$, the matrix

$$\tilde{A}(c) = \begin{bmatrix} A_{c[0]} & -I & 0 & \dots & 0 \\ 0 & A_{c[1]} & -I & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -I & 0 & 0 & \dots & A_{c[\kappa-1]} \end{bmatrix} \quad (11)$$

and $\tilde{b}(c) = [b'_{c[0]} \ b'_{c[1]} \ \dots \ b'_{c[\kappa-1]}]'$. Notice that the boundary condition $x_e[0] = x_e[\kappa]$ has been taken into account in the last row of $\tilde{A}(c)$ and that due to (3) all the limit cycles satisfy the identity $\ell_{c[n]}[n] = 0$.

The N^κ different switching sequences $c \in \mathcal{C}(\kappa)$ allow us to define a family of candidate limit cycles

$$\mathfrak{X} = \{\mathcal{X}_e(c) : c \in \mathcal{C}(\kappa)\} \quad (12)$$

The search for a desired limit cycle $\mathcal{X}_e^* \in \mathfrak{X}$, defining properties of interest for the trajectories of (1)–(2) in the steady-state, as ripple amplitude and oscillation frequency, can be constrained to a subset $\mathfrak{X}_s \subseteq \mathfrak{X}$, which is related to some criterion specified by the designer. Given a reference point x_* , a possible criterion is

$$\mathfrak{X}_s = \left\{ \mathcal{X}_e \in \mathfrak{X} : \frac{1}{\kappa} \sum_{n=0}^{\kappa-1} \|\Gamma(x_e[n] - x_*)\| < 1 \right\} \quad (13)$$

which contains candidate limit cycles whose mean distance between $\Gamma x_e[n]$ and Γx_* over $n \in \{0, \dots, \kappa - 1\}$ is smaller than 1. The matrix Γ , provided by the designer, is important when the interest is to optimise the steady-state behaviour of only one or a combination of state components. Alternatively, when the goal is to bound the ripple amplitudes of the trajectories in the steady-state, another possibility is to adopt the subset

$$\mathfrak{X}_s = \left\{ \mathcal{X}_e \in \mathfrak{X} : \max_{n \in \{0, \dots, \kappa-1\}} \|\Gamma(x_e[n] - x_*)\|_\infty < 1 \right\} \quad (14)$$

with, as in the first case, Γ being a parameter provided by the designer. Notice that the criterion choice depends exclusively on the interest of the designer with respect to the steady-state behaviour of the trajectories, which can define different subsets $\mathfrak{X}_s$. The subset of sequences c associated to $\mathcal{X}_e(c) \in \mathfrak{X}_s$ is defined as $\mathcal{C}_s(\kappa) \subseteq \mathcal{C}(\kappa)$. The next theorem relies upon the results of [Bittanti and Colaneri \(2009\)](#) to provide conditions that assure the existence of a globally asymptotically stable limit cycle $\mathcal{X}_e(c) \in \mathfrak{X}_s$ to be considered in the forthcoming developments.

Theorem 1. Consider system (1)–(2) with $w[n] = 0$, $\forall n \in \mathbb{N}_-$, and let $0 < \kappa \in \mathbb{N}$ and $c \in \mathcal{C}_s(\kappa)$ be given. The limit cycle $\mathcal{X}_e(c) \in \mathfrak{X}_s$ under the periodic switching sequence $\sigma[n] = c[k(n)]$ is globally asymptotically stable if and only if there exist positive definite matrices $P[n] > 0$, $n \in \{0, \dots, \kappa - 1\}$, satisfying the linear matrix inequalities

$$A'_{c[n]}P[n+1]A_{c[n]} - P[n] < 0 \quad (15)$$

for all $n \in \{0, \dots, \kappa - 1\}$, with the boundary condition $P[\kappa] = P[0]$.

Proof. Consider system (1)–(2) with $w[n] = 0$, $\forall n \in \mathbb{N}_-$, written alternatively as (5)–(6), and governed by the periodic switching sequence $\sigma[n] = c[k(n)] \in \mathcal{C}_s(\kappa)$. Associated to this sequence, the limit cycle $\mathcal{X}_e(c) \in \mathfrak{X}_s$ defined in (4) assures that $A(c)\tilde{x}_e = -\tilde{b}(c)$ holds, which is equivalent to verify the identity

$$\ell_{c[n]}[n] = A_{c[n]}x_e[n] - x_e[n+1] + b_{c[n]} = 0 \quad (16)$$

for all $n \in \{0, \dots, \kappa - 1\}$ and $x_e[0] = x_e[\kappa]$. With the identity (16), the system becomes linear $\xi[n+1] = A_{c[n]}\xi[n]$. Hence, due to Proposition 3.5 of [Bittanti and Colaneri \(2009\)](#), the condition (15) is necessary and sufficient for global asymptotic stability of the origin $\xi = 0$ regarding system (5)–(6), or equivalently, for the global asymptotic stability of the limit cycle $\mathcal{X}_e(c)$ concerning system (1)–(2). $\square$

For a desired limit cycle $\mathcal{X}_e^*$, chosen inside $\mathfrak{X}_s$, that satisfies the conditions of [Theorem 1](#), it is possible to determine a state-dependent switching function that not only preserves global asymptotic stability of the limit cycle, but also optimises an upper bound for the previously presented $\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance indexes. Before proceeding, let us define the matrix function

$$\mathcal{L}_i[n] = \begin{bmatrix} A'_iP[n+1]A_i - P[n] & \bullet \\ \ell_i[n]'P[n+1]A_i & \ell_i[n]'P[n+1]\ell_i[n] \end{bmatrix} \quad (17)$$

since it will be useful afterwards.

3.2. Stability and guaranteed cost

Let us consider a simpler switched affine system given by

$$\xi[n+1] = A_{\sigma[n]}\xi[n] + \ell_{\sigma[n]}[n] \quad (18)$$

$$z_e[n] = E_{\sigma[n]}\xi[n] \quad (19)$$

defined for $n \in \mathbb{N}$ and evolving from an arbitrary initial condition $\xi[0] = x[0] - x_e[0]$ with $x_e[n] \in \mathcal{X}_e^*$. At this first moment, our goal is to determine $\sigma[n] = u(x[n])$ that preserves global asymptotic stability of the limit cycle $\mathcal{X}_e^* \in \mathfrak{X}_s$ chosen as the one that minimises the upper bound for the square L_2 -norm $\|z_e\|_2^2$. The next theorem presents this result.

Theorem 2. Consider system (1)–(2) defined for $n \in \mathbb{N}$ with $w[n] = 0$, $\forall n \in \mathbb{N}$ and evolving from an arbitrary $x[0]$. Let the scalar $0 < \kappa \in \mathbb{N}$ and the subset of candidate limit cycles $\mathcal{X}_e(c) \in \mathfrak{X}_s$, $\forall c \in \mathcal{C}_s(\kappa)$, be given. If there exist positive definite matrices $P[n] > 0$ satisfying the optimisation problem

$$\min_{\mathcal{X}_e(c) \in \mathfrak{X}_s} \inf_{P[n]} (x[0] - x_e[0])'P[0](x[0] - x_e[0]) \quad (20)$$

subject to the linear matrix inequalities

$$A'_{c[n]}P[n+1]A_{c[n]} - P[n] + E'_{c[n]}E_{c[n]} < 0 \quad (21)$$

for all $n \in \{0, \dots, \kappa - 1\}$, $c \in \mathcal{C}_s(\kappa)$ with the boundary condition $P[\kappa] = P[0]$, then the state-dependent switching function $\sigma[n] = u(\xi[n])$ with

$$u(\xi) = \arg \min_{i \in \mathbb{K}} \begin{bmatrix} \xi \\ 1 \end{bmatrix}' \mathcal{L}_i[k(n)] \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' E'_i E_i \xi \quad (22)$$

assures that the limit cycle $\mathcal{X}_e^* = \mathcal{X}_e^*(c)$, solution of (20), is globally asymptotically stable and that the following upper bound

$$\|z_e\|_2^2 < (x[0] - x_e[0])' P[0] (x[0] - x_e[0]) \quad (23)$$

is a guaranteed cost of performance.

Proof. Notice that the system under consideration can be rewritten alternatively as (18)–(19) and adopt the switching function (22). For an arbitrary trajectory of (18)–(19) in the time interval $n \in \{0, \dots, \kappa - 1\}$, the Lyapunov function (9) provides

$$\begin{aligned} \Delta v(\xi, n) &= \begin{bmatrix} \xi \\ 1 \end{bmatrix}' \mathcal{L}_{\sigma[n]}[n] \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' E'_{\sigma[n]} E_{\sigma[n]} \xi - z_e' z_e \\ &= \min_{i \in \mathbb{K}} \begin{bmatrix} \xi \\ 1 \end{bmatrix}' \mathcal{L}_i[n] \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' E'_i E_i \xi - z_e' z_e \\ &\leq \begin{bmatrix} \xi \\ 1 \end{bmatrix}' \mathcal{L}_{c[n]}[n] \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' E'_{c[n]} E_{c[n]} \xi - z_e' z_e \\ &= \xi' (A'_{c[n]}P[n+1]A_{c[n]} - P[n] + E'_{c[n]}E_{c[n]}) \xi - z_e' z_e \\ &< -z_e' z_e \end{aligned} \quad (24)$$

where the second equality comes from the switching function (22), the first inequality is a consequence of the minimum operator and the third equality is due to the fact that the sequence $c \in \mathcal{C}_s(\kappa)$, associated with the limit cycle $\mathcal{X}_e(c) \in \mathcal{X}_s$, assures that the identity (16) holds. Finally, the last inequality is verified by the validity of (21). From the periodic continuation $P[n] = P[k(n)]$ we have that $\Delta v(\xi, n) < -z_e' z_e < 0$ for all $n \in \mathbb{N}$ and, therefore, the equilibrium point $\xi = 0$ is globally asymptotically stable. Consequently, the same occurs for the limit cycle $\mathcal{X}_e^*$. Moreover, summing both sides of (24) from $n = 0$ up to infinity, we obtain a telescoping series, which assures

$$\sum_{n=0}^{\infty} \Delta v(\xi[n], n) = -v(\xi[0], 0) < -\sum_{n \in \mathbb{N}} \|z_e[n]\|^2 \quad (25)$$

verifying (23) and concluding the proof. $\square$

This theorem presents conditions for the design of the mentioned state-dependent switching function. Notice that, for the particular case where $\sigma[n] = c[k(n)]$, it is simple to show that the equality $\|z_e\|_2^2 = (x[0] - x_e[0])' P[0] (x[0] - x_e[0])$ holds, where $P[0] > 0$ is the solution of $A'_{c[n]}P[n+1]A_{c[n]} - P[n] + E'_{c[n]}E_{c[n]} = 0$ for all $n \in \{0, \dots, \kappa - 1\}$ and $P[\kappa] = P[0]$. This indicates that the state-dependent switching function (22) provides an actual cost which is better than, or at least equal to, the one provided by a periodic switching function.

To the best of the authors' knowledge, this is the first result dealing with switching function design assuring global asymptotic stability of a desired limit cycle for discrete-time switched affine systems. Even in the continuous-time domain there are only few results in the literature, as for instance, the recent reference (Benmiloud et al., 2019) that deals with local asymptotic stability by means of a systematic methodology based on a Poincaré map approach. Differently, the conditions proposed in Theorem 2 are expressed by means of LMIs being, therefore, simple to solve by readily available algorithms. Compared to

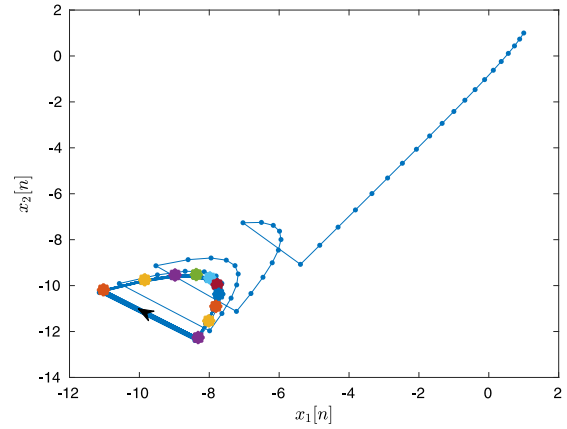


Fig. 1. State trajectory $x[n]$ and limit cycle $\mathcal{X}_e^*$.

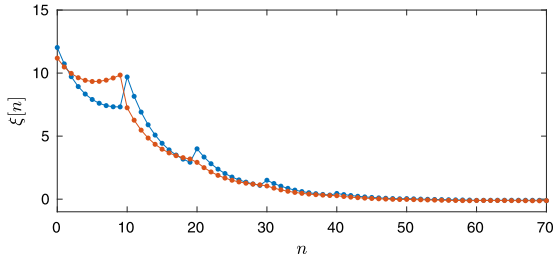
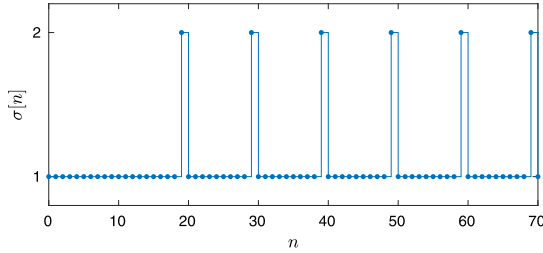
the conditions for practical stability, the inequalities (21) are less conservative than the methods based on a simple time-invariant quadratic Lyapunov function as in Deaecto and Geromel (2017), Deaecto and Santos (2015) and are not comparable to the Lyapunov–Metzler conditions available in Egidio and Deaecto (2019), as discussed in Daiha et al. (2017). The following example illustrates this result.

Example 1. Consider the switched affine system (1)–(2) defined for $n \in \mathbb{N}$ with $w[n] = 0$, $\forall n \in \mathbb{N}$ and composed of two unstable subsystems given by $A_i = e^{A_{ci}T}$, $b_i = \int_0^T e^{A_{ci}\tau} d\tau b_{ci}$, $\forall i \in \mathbb{K}$, with $T = 0.1[s]$ and

$$A_{c1} = \begin{bmatrix} -4 & 3 \\ -3 & 2.5 \end{bmatrix}, A_{c2} = \begin{bmatrix} 4 & -1 \\ 1 & -2 \end{bmatrix}, b_{c1} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, b_{c2} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

and $E_1 = E_2 = I$. The goal is to maintain the first state component around a mean value near -9 when in the steady-state. To this end, for $\kappa = 10$, a subset of candidate limit cycles was defined as in (13) with $x_* = [-9 \ ?]'$, where the symbol “?” indicates a value without importance, $\Gamma = [1 \ 0]$, which produced an $\mathcal{X}_s \subset \mathcal{X}$ with 10 candidate limit cycles whose mean value of the first state component is inside the interval $(-10, -8)$. Solving the conditions of Theorem 2, matrices $P[n]$ and the optimal limit cycle $\mathcal{X}_e^*$, correspondent to the optimal sequence $c = (1, 1, 1, 1, 1, 1, 1, 1, 1, 2) \in \mathcal{C}_s(\kappa)$, have been found, allowing the implementation of the switching function (22). The limit cycle $\mathcal{X}_e^*$ is presented in Fig. 1 along with a trajectory $x[n]$ starting from $x[0] = [1 \ 1]'$, where the arrow represents the direction of the state trajectories on $\mathcal{X}_e^*$. Fig. 2 and Fig. 3 show, respectively, the state trajectories $\xi[n]$ and the switching signal $\sigma[n]$ over time instants n . Notice that the trajectories $x[n]$ converged to the chosen limit cycle whilst, as expected, $\xi[n]$ approached zero. Moreover, by numerical simulation we have calculated $\|z\|_2^2 = 2417.53$ which respects its upper bound 3945.57, defined in (23). This puts in evidence the efficiency of the proposed switching strategy.

About this result, an interesting remark is that a better performance can be obtained by a suitable choice of κ , allowing the performance optimisation in the steady and transient states. Fortunately, $\kappa > 0$ does not need to be large for obtaining a suitable guaranteed cost. Actually, as it has been discussed in Deaecto and Geromel (2018), whenever $\hat{\kappa} = \eta\kappa$, for some $0 < \eta \in \mathbb{N}$, it is possible to assure that the costs are non increasing as η increases, since $\mathcal{C}_s(\eta\kappa)$ contains all the periodic sequences $\mathcal{C}_s(\kappa)$ as particular case. However, nothing can be concluded about the costs for the case when $\hat{\kappa} > \kappa$ but $\hat{\kappa} \neq \eta\kappa$ for $2 \leq \eta \in \mathbb{N}$.

Fig. 2. Time evolution of the auxiliary state $\xi[n]$.Fig. 3. Obtained switching signal $\sigma[n]$.

3.3. $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control design

At this point, let us generalise Theorem 2 to cope with $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control design, whose indexes have been defined in (7) and (8), respectively. Consider the more general switched affine system (5)–(6) with external input $w[n] = e_r \delta[n+1]$, where $e_r, r \in \{0, \dots, n_w\}$, is the standard basis for $\mathbb{R}^{n_w}$. This system can be equivalently rewritten as (18)–(19) but evolving from initial condition $\xi[0] = \ell_{\sigma[-1]}[-1] + H_{\sigma[-1]}e_r$. The next corollary presents an upper bound for the $\mathcal{H}_2$ performance index.

Corollary 1. Consider system (1)–(2) defined for $n \in \mathbb{N}_-$ with $w[n] = e_r \delta[n+1]$. Let the scalar $0 < \kappa \in \mathbb{N}$, the subset of candidate limit cycles $\mathcal{X}_e(c) \in \mathcal{X}_s, \forall c \in \mathcal{C}_s(\kappa)$, and $\sigma[-1] = m$ be given. If there exist positive definite matrices $P[n] > 0$ satisfying the optimisation problem

$$\min_{\mathcal{X}_e(c) \in \mathcal{X}_s} \inf_{P[n]} \text{tr}((L_m + H_m)'P[0](L_m + H_m) + G_m'G_m) \quad (26)$$

where $L_m = [\ell_m[-1] \dots \ell_m[-1]] \in \mathbb{R}^{n_x \times n_w}$ and $\ell_m[-1] = A_m x_e[-1] - x_e[0] + b_m$, subject to the linear matrix inequalities (21), for all $n \in \{0, \dots, \kappa - 1\}$, $c \in \mathcal{C}_s(\kappa)$ with the boundary condition $P[\kappa] = P[0]$, then the state-dependent switching function $\sigma[n] = u(\xi[n])$ given in (22) assures that the limit cycle $\mathcal{X}_e^* = \mathcal{X}_e(c)$, solution of (26), is globally asymptotically stable and the upper bound

$$J_2(\sigma) < \text{tr}((L_m + H_m)'P[0](L_m + H_m) + G_m'G_m) \quad (27)$$

is an $\mathcal{H}_2$ guaranteed cost of performance.

Proof. From Theorem 2, asymptotic stability is assured by the switching function (22) and the inequality $\|z_e\|_2^2 < \xi[0]'P[0]\xi[0]$ with $\xi[0] = x[0] - x_e[0]$ holds. Hence, from (7), we have

$$\begin{aligned} J_2(\sigma) &= \sum_{r=1}^{n_w} \|z_{er}\|_2^2 + e_r' G_m' G_m e_r \\ &< \sum_{r=1}^{n_w} \xi[0]' P[0] \xi[0] + e_r' G_m' G_m e_r \\ &= \text{tr}((L_m + H_m)' P[0] (L_m + H_m) + G_m' G_m) \end{aligned} \quad (28)$$

where the inequality comes from the guaranteed cost of Theorem 2 and the last equality is obtained from $\xi[0] = (L_m + H_m)e_r$. The proof is concluded. $\square$

Notice that, for a given subset of candidate limit cycles $\mathcal{X}_s$ associated to $c \in \mathcal{C}_s(\kappa) \subseteq \mathcal{C}(\kappa)$, Corollary 1 requires the solution of a certain number of convex subproblems that does not overcome N^κ , and corresponds to the number of elements of $\mathcal{X}_s$. Making the optimisation with respect to $c \in \mathcal{C}_s(\kappa)$, we obtain the optimal limit cycle $\mathcal{X}_e^* \in \mathcal{X}_s$, which minimises the $\mathcal{H}_2$ guaranteed cost, whilst jointly design the switching function.

Turning our attention to the $\mathcal{H}_\infty$ control design, let us consider again system (5)–(6) but defined for $n \in \mathbb{N}$, with external input $w \in \mathcal{L}_2$ and $\xi[0] = 0$. Defining $\mathcal{E}_i[n] = H_i' P[n+1] H_i + G_i' G_i - \rho I$, the next matrix function

$$\begin{aligned} \mathcal{M}_i[n] &= \begin{bmatrix} A_i' P[n+1] H_i + E_i' G_i \\ \ell_i[n]' P[n+1] H_i \end{bmatrix} \mathcal{E}_i[n+1]^{-1} \times \\ &\times \begin{bmatrix} A_i' P[n+1] H_i + E_i' G_i \\ \ell_i[n]' P[n+1] H_i \end{bmatrix}' \end{aligned} \quad (29)$$

will be very useful to obtain the conditions for the $\mathcal{H}_\infty$ control design, as it will be clear in the next corollary.

Corollary 2. Consider system (1)–(2) defined for $n \in \mathbb{N}$ with $w \in \mathcal{L}_2$, evolving from $x[0] = x_e[0]$. Let the scalar $0 < \kappa \in \mathbb{N}$ and the subset of candidate limit cycles $\mathcal{X}_e(c) \in \mathcal{X}_s, \forall c \in \mathcal{C}_s(\kappa)$ be given. If there exist matrices $P[n] > 0$, a scalar $\rho > 0$, solution of the optimisation problem

$$\min_{\mathcal{X}_e(c) \in \mathcal{X}_s} \inf_{P[n], \rho} \rho \quad (30)$$

subject to the linear matrix inequalities

$$\begin{bmatrix} P[n] & \bullet & \bullet & \bullet \\ 0 & \rho I & \bullet & \bullet \\ P[n+1] A_{c[n]} & P[n+1] H_{c[n]} & P[n+1] & \bullet \\ E_{c[n]} & G_{c[n]} & 0 & I \end{bmatrix} > 0 \quad (31)$$

$$H_i' P[n+1] H_i + G_i' G_i - \rho I < 0, \quad i \in \mathbb{K} \quad (32)$$

for all $n \in \{0, \dots, \kappa - 1\}$, $c \in \mathcal{C}_s(\kappa)$ with $P[\kappa] = P[0]$, then the switching function $\sigma[n] = u(\xi[n])$ with

$$u(\xi) = \arg \min_{i \in \mathbb{K}} \begin{bmatrix} \xi \\ 1 \end{bmatrix}' (\mathcal{L}_i[k(n)] - \mathcal{M}_i[k(n)]) \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' E_i' E_i \xi \quad (33)$$

assures that the limit cycle $\mathcal{X}_e^* = \mathcal{X}_e(c)$, solution of (30), is globally asymptotically stable and verifies the inequality $J_\infty(\sigma) < \rho$.

Proof. Consider system (1)–(2) written alternatively as (5)–(6) with $\xi[0] = 0$, denote $\xi[n] = \xi$, $\sigma[n] = \sigma$ and $w[n] = w$, and define

$$\mathcal{F}_i[n] = \begin{bmatrix} Q_i[n] & \bullet & \bullet \\ \ell_i[n]' P[n+1] A_i & \ell_i[n]' P[n+1] \ell_i[n] & \bullet \\ H_i' P[n+1] A_i + G_i' E_i & H_i' P[n+1] \ell_i[n] & \mathcal{E}_i[n] \end{bmatrix}$$

with $Q_i[n] = A_i' P[n+1] A_i - P[n] + E_i' E_i$, as well as the augmented variable $\tilde{\xi} = [\xi' \ 1 \ w']'$. Adopting the Lyapunov function (9), we have, inside the time interval $n \in \{0, \dots, \kappa - 1\}$, the following developments

$$\begin{aligned} \Delta v(\xi, n) &= \tilde{\xi}' \mathcal{F}_\sigma[n] \tilde{\xi} - z_e' z_e + \rho w' w \\ &\leq \min_{i \in \mathbb{K}} \begin{bmatrix} \xi \\ 1 \end{bmatrix}' (\mathcal{L}_i[n] - \mathcal{M}_i[n]) \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' Q_i \xi - z_e' z_e + \rho w' w \\ &\leq \begin{bmatrix} \xi \\ 1 \end{bmatrix}' (\mathcal{L}_{c[n]}[n] - \mathcal{M}_{c[n]}[n]) \begin{bmatrix} \xi \\ 1 \end{bmatrix} + \xi' Q_{c[n]} \xi - z_e' z_e + \rho w' w \\ &= \xi' (A_{c[n]}' P[n+1] A_{c[n]} - P[n] + Q_{c[n]}) \xi \end{aligned}$$

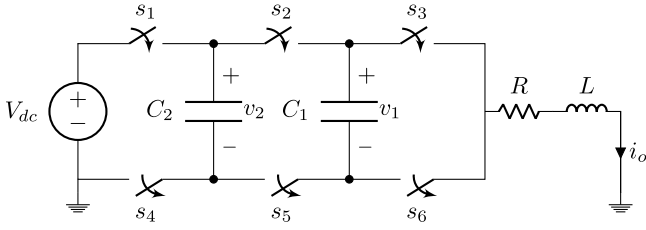


Fig. 4. Schematic of a three-cell converter.

$$\begin{aligned} & -\mathcal{N}_i[n]\mathcal{E}_{c[n]}[n]^{-1}\mathcal{N}_i[n']\xi - z_e'z_e + \rho w'w \\ & < -z_e'z_e + \rho w'w \end{aligned} \quad (34)$$

with $\mathcal{N}_i[n] = A_i'P[n+1]H_i + E_i'G_i$ and $Q_i = E_i'E_i$. The first inequality comes from the fact that, due to (32) the function $f_i(\xi, w) = \xi'F_i[n]\xi$ is concave with respect to w for all $i \in \mathbb{K}$. Hence, it is possible to determine $\sup_{w \in \mathcal{L}_2} f_\sigma(\xi, w)$ which occurs for

$$w^* = -\mathcal{E}_\sigma[n]^{-1}(\mathcal{N}_\sigma[n]\xi + H_\sigma'P[n+1]\ell_\sigma[n]) \quad (35)$$

This value together with the switching function provided in (33) results in the expression of the right hand side of the first inequality. The second inequality is a consequence of the minimum operator, the second equality is due to the fact that $\ell_{c[n]}[n] = 0$, since the sequence $c \in \mathcal{C}_s(\kappa)$ is associated to the limit cycle $\mathcal{X}_e(c) \in \mathcal{X}_s$. Finally, the last inequality comes from the validity of (31). The periodic continuation $P[n] = P[k(n)]$ assures that $\Delta v(\xi, n) < -z_e'z_e + \rho w'w$ for all $n \in \mathbb{N}$. Summing both sides of this inequality from $n = 0$ up to infinity, and recalling that $v(\xi, 0) = 0$ since $\xi[0] = 0$ and $\lim_{n \rightarrow \infty} v(\xi, n) = 0$ as a consequence of the asymptotic stability of the origin $\xi = 0$, we obtain

$$\|z_e\|_2^2 - \rho \|w\|_2^2 < 0 \quad (36)$$

which assures the validity of the $\mathcal{H}_\infty$ guaranteed cost, concluding thus the proof. $\square$

The results we have just presented treat the control of discrete-time switched affine systems considering the performance optimisation in the steady and transient states. More specifically, we have presented the switching function design for this class of systems in two sequential steps. First, we have obtained a subset of candidate limit cycles $\mathcal{X}_e(c) \in \mathcal{X}_s$ based on some criterion chosen by the designer. As a second step, we have determined a state-dependent switching function in order to assure global asymptotic stability of the limit cycle $\mathcal{X}_e^* \in \mathcal{X}_s$ correspondent to the least $\mathcal{H}_2$ or $\mathcal{H}_\infty$ guaranteed cost through Corollary 1 or 2, respectively.

Another remark is that, considering one of the criteria (13) or (14), whenever $\mathcal{X}_s$ is non-empty it has at least κ candidate limit cycles, which are shifted versions of each other. As an example, the sequences $c \in \{(1, 2, 2, 1), (2, 2, 1, 1), (2, 1, 1, 2), (1, 1, 2, 2)\}$ provide the same limit cycle but with different initial points $x_e[0]$. Therefore, if one of them is inside $\mathcal{C}_s(\kappa)$ the others are as well. Whenever this scenario takes place in the $\mathcal{H}_2$ control design, the optimisation with respect to $\mathcal{X}_e(c) \in \mathcal{X}_s$ consists in determining the best initial value $x_e[0]$ such that the guaranteed cost is minimised. On the other hand, the $\mathcal{H}_\infty$ design procedure produces the same optimal upper bound ρ for all the shifted versions since the optimisation problem in Corollary 2 is independent of $x_e[0]$. The next practical application example puts in evidence the validity and efficiency of the proposed theory.

Example 2. This example was borrowed from Benmiloud et al. (2019), which also treats the asymptotic stability of a limit cycle,

Table 1

Values u_1 , u_2 , and u_3 for each operation mode i .

i	1	2	3	4	5	6	7	8
u_1	0	1	0	1	0	1	0	1
u_2	0	0	1	1	0	0	1	1
u_3	0	0	0	0	1	1	1	1

but in the continuous-time domain and guaranteeing only local stability. It considers a multicellular DC-DC converter feeding an inductive load as depicted in Fig. 4, which is composed of a series association of elementary commutation cells, each one consisting of a pair of complementary switches (i.e., when s_1 is open, s_4 is closed). The system parameters are $V_{dc} = 60$ [V], $C_1 = C_2 = 40$ [μ F], $L = 5$ [mH], and $R = 20$ [Ω]. This topology presents $N = 8$ operation modes (or subsystems) from which a switching signal $\sigma(t)$ selects one of them as active at each instant of time. This system can be modelled as the continuous-time switched affine system

$$\dot{x}(t) = A_{c\sigma(t)}x(t) + b_{c\sigma(t)} + H_{c\sigma(t)}w(t) \quad (37)$$

$$z_c(t) = E_{c\sigma(t)}x(t) + G_{c\sigma(t)}w(t) \quad (38)$$

where the state is defined as $x(t) = [v_1(t) \ v_2(t) \ i_o(t)]'$, the exogenous input $w(t)$ represents a disturbance and $\sigma(t)$ must be governed by a switching function to be designed. The system matrices are

$$A_{ci} = \begin{bmatrix} 0 & 0 & \frac{u_2 - u_1}{C_1} \\ 0 & 0 & \frac{u_3 - u_2}{C_2} \\ \frac{u_1 - u_2}{L} & \frac{u_2 - u_3}{L} & -\frac{R}{L} \end{bmatrix}, \quad b_{ci} = \begin{bmatrix} 0 \\ 0 \\ \frac{V_{dc}u_3}{L} \end{bmatrix}, \quad (39)$$

$E_{ci} = \text{diag}([1 \ 1 \ 1])$ and $G_{ci} = [0 \ 0 \ 0]'$. The matrices $H_{ci} \in \mathbb{R}^3$, $\forall i \in \mathbb{K}$, will be defined afterwards. The values for u_1 , u_2 , and u_3 are given in Table 1 as a function of the current value of the switching signal $\sigma(t) = i$. In order to treat this problem in the discrete-time domain, assuring an upper bound for the switching frequency, a sampling period $T = 0.1$ [ms] is taken into account, such that the switching signal must respect $\sigma(t) = \sigma(t_n)$ for all $t \in [t_n, t_{n+1})$ where $t_n = nT$ is the n th sampling instant. Denoting $x(t_n) = x[n]$, $\forall n \in \mathbb{N}$, let us consider the exact discretisation procedure presented in Souza, Deaecto, Geromel, and Daafouz (2014), which provides a controlled output $z[n]$ that satisfies

$$\|z[n]\|_2^2 = \|z_c(t)\|_2^2 = \int_0^\infty z_c(t)'z_c(t)dt \quad (40)$$

indicating that L_2 -norms of the controlled outputs $z_c(t)$ and $z[n]$ are identical. With this procedure, we can obtain the discrete-time switched affine system (1)–(2) whose matrices are given by

$$\begin{bmatrix} A_i & [b_i \ H_i] \\ 0 & I \end{bmatrix} = e^{A_i T}, \quad \begin{bmatrix} E_i' \\ G_i' \end{bmatrix} \begin{bmatrix} E_i' \\ G_i' \end{bmatrix}' = \int_0^T e^{A_i' t} C_i' C_i e^{A_i t} dt \quad (41)$$

with

$$\mathcal{A}_i = \begin{bmatrix} A_{ci} & [b_{ci} \ H_{ci}] \\ 0 & 0 \end{bmatrix}, \quad C_i = [E_{ci} \ G_{ci}]$$

for all $i \in \mathbb{K}$, where it is assumed that the external input $w(t) = w(t_n)$, $\forall t \in [t_n, t_{n+1})$, is piecewise constant, which can be a good approximation for a low frequency disturbance $w(t)$.

Hence, concerning the discretised system, our goal is to design a limit cycle $\mathcal{X}_e^*$ and a switching function $\sigma[n] = u(\xi[n])$ able to govern the state trajectories $x[n]$ from any initial condition towards $\mathcal{X}_e^*$, assuring $\mathcal{H}_2$ or $\mathcal{H}_\infty$ performance indexes. Each case will be treated separately in the sequel. In contrast with the results of Benmiloud et al. (2019), which guarantee only local stability, the proposed methodology assures global asymptotic stability of the limit cycle $\mathcal{X}_e^*$.

Before proceeding to the control design, the subset of candidate limit cycles has to be determined. The limit cycle $\mathcal{X}_e^*$ must be chosen in order to keep a suitable maximum distance to the reference point $x_* = [V_{dc}/3 \ 2V_{dc}/3 \ I_{ref}]$, with a free $I_{ref} \in [0, I_{max}]$, where $I_{max} = V_{dc}/R$ is the maximum possible current at the output of the converter. Notice that x_* is not an equilibrium point of any subsystem and its asymptotic stability is impossible in the discrete-time domain. Our goal is accomplished by taking $\kappa = 6$ and adopting (14) with $\Gamma = \text{diag}\{0.5, 0.5, 0\}$ in order to construct the subset of candidate limit cycles $\mathcal{X}_s$. Using these parameters, we have obtained 30 candidates from which the one providing the best transient response is chosen. The given Γ imposes that the voltages v_1 and v_2 must not deviate more than 2 [V] from the references specified in x_* during the steady-state.

$\mathcal{H}_2$ control design: Our interest is to design a switching function $\sigma[n] = u(\xi[n])$ in order to optimise the transient response of the discretised system (1)–(2), starting from a null initial condition $x[0] = 0$, which represents the system start-up. Considering that in the $\mathcal{H}_2$ case, the external input is not piecewise constant, but of impulsive type, matrices H_i , $\forall i \in \mathbb{K}$, are determined differently from those provided in the discretisation procedure. Indeed, for $w[n] = \delta[n+1]$, $x[-1] = x_e[-1]$ and choosing $\sigma[-1] = c[\kappa-1] = m$, the discrete-time matrices are given by $H_i = -x_e[0]$, $\forall i \in \mathbb{K}$, which lead to $x[0] = 0$. Under these conditions, solving the optimisation problem of Corollary 1, we have obtained an $\mathcal{H}_2$ guaranteed cost of $J_2 < 37.7903$ associated to the sequence $c = (5, 1, 3, 1, 2, 1)$ and the optimal limit cycle $\mathcal{X}_e^*$ defined by the fundamental period

$$(x_e[0], \dots, x_e[5]) = \left(\begin{bmatrix} 19.5989 \\ 39.4582 \\ 0.4031 \end{bmatrix}, \begin{bmatrix} 19.5989 \\ 40.7315 \\ 0.5983 \end{bmatrix}, \begin{bmatrix} 19.5989 \\ 20.8722 \\ 0.5961 \end{bmatrix}, \begin{bmatrix} 20.8722 \\ 39.4582 \\ 0.3996 \end{bmatrix}, \begin{bmatrix} 39.4582 \\ 19.5989 \\ 0.6014 \end{bmatrix}, \begin{bmatrix} 39.4582 \\ 40.7315 \\ 0.4011 \end{bmatrix} \right) \quad (42)$$

starting from $n = 0$. Implementing the switching function $\sigma[n] = u(\xi[n])$, given in (22), Fig. 5 presents the state trajectories and Fig. 6 shows the corresponding switching signal obtained by the discrete-time simulation. A phase portrait of the system trajectories towards the limit cycle is given in Fig. 7 and a zoom on the limit cycle under the steady-state operation is presented in Fig. 8, where the arrow represents the direction of the state trajectory. Evaluating numerically the $\mathcal{H}_2$ performance index (7) we have obtained $J_2 = 1.7632$, which shows the efficiency of the designed switching function, since by adopting the periodic switching function $\sigma[n] = c[k(n)]$ the value $J_2 = 37.7903$ is obtained.

$\mathcal{H}_\infty$ control design: Now our goal is to design a switching function $\sigma[n] = u(\xi[n])$ but to attenuate the influence of an exogenous input representing a voltage oscillation and a voltage dip of the input source V_{dc} . Voltage dips (or voltage sags) are recurrent events that depreciate power quality and may generate failures in sensitive loads such as medical equipment, factory automations, among others, see Dargahi, Sadigh, Pahlavani, and Shoulaie (2012).

Considering now the continuous-time matrices $H_{ci} = [0 \ 0 \ u_3/L]'$, $\forall i \in \mathbb{K}$, which model $w(t)$ as a deviation of the input voltage around the considered V_{dc} . Taking into account the discretised system (1)–(2) with matrices given in (41), a limit cycle $\mathcal{X}_e^* \in \mathcal{X}_s$ must now be chosen in order to minimise ρ . Solving the optimisation problem of Corollary 2, we have assured an $\mathcal{H}_\infty$ guaranteed cost of $J_\infty < 0.3633 \times 10^{-3}$ associated to the sequence $c = (3, 1, 2, 1, 5, 1)$ and the correspondent optimal limit cycle $\mathcal{X}_e^*$, which is the same of the $\mathcal{H}_2$ case but shifted by 2. Implementing the switching function $\sigma[n] = u(\xi[n])$ with $u(\xi[n])$ given in (33),

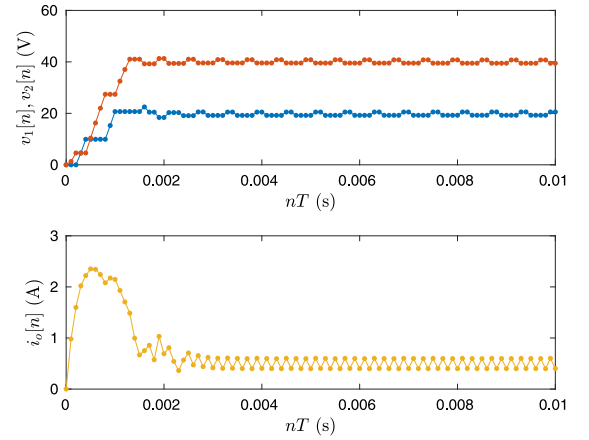


Fig. 5. State trajectories for $\mathcal{H}_2$ control design.

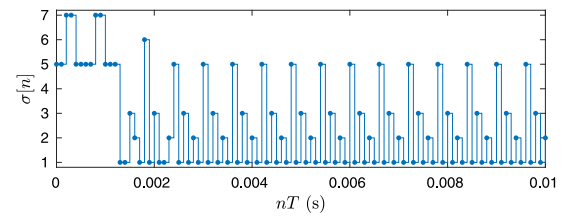


Fig. 6. Obtained switching signal for $\mathcal{H}_2$ control design.

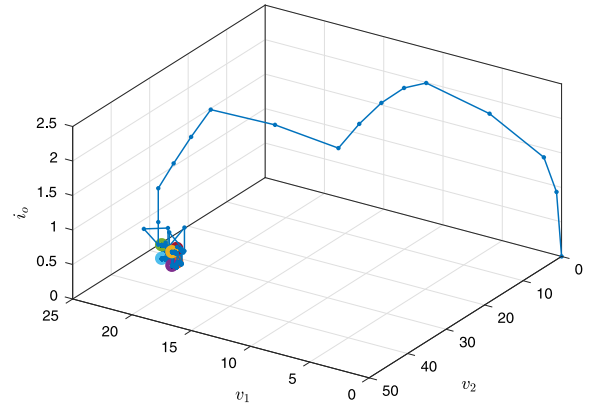


Fig. 7. State trajectories converging to the limit cycle $\mathcal{X}_e^*$.

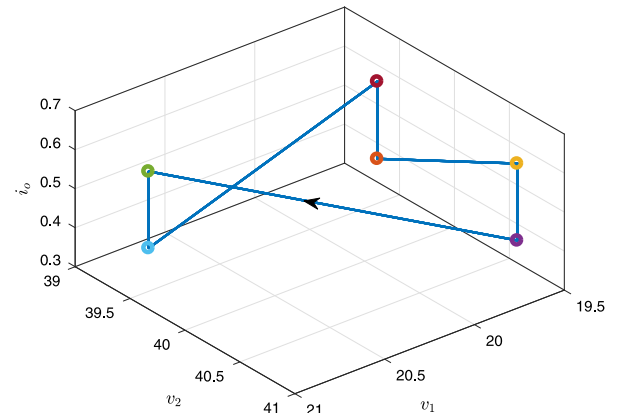


Fig. 8. Zoom showing the limit cycle $\mathcal{X}_e^*$ in steady-state.

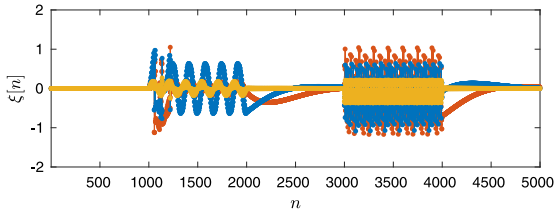


Fig. 9. Discrete-time state $\xi[n]$ for $\mathcal{H}_\infty$ control design.

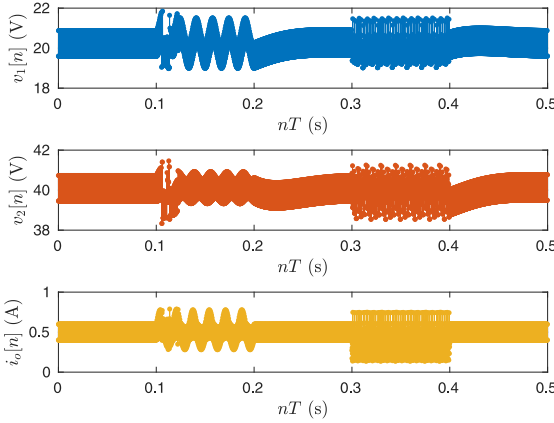


Fig. 10. State trajectories for $\mathcal{H}_\infty$ control design.

starting the system from $x[0] = [19.5989 \ 40.7315 \ 0.4011]' = x_e[0]$ and considering the following exogenous input

$$w[n] = \begin{cases} 10 \sin(120\pi Tn), & n \in [0.1/T, 0.2/T) \\ -20, & n \in [0.3/T, 0.4/T) \\ 0, & \text{otherwise} \end{cases} \quad (43)$$

we have obtained a discrete-time state trajectories $\xi[n]$ and $x[n]$ depicted in Figs. 9 and 10, respectively. Moreover, the RMS value of the output current $i_o[n]$ has been calculated over three different intervals, namely voltage oscillation $n \in [0.1/T, 0.2/T)$, voltage dip $n \in [0.3/T, 0.4/T)$ and regular operation $n \in [0, 0.1/T)$ providing 0.5161 [A], 0.4231 [A] and 0.5096 [A], respectively. This demonstrates that the designed switching function was successful in reducing the influence of the disturbance $w[n]$ on the system behaviour, guaranteeing an RMS deviation of approximately 17% during a voltage dip of 33%, compared to the regular operation.

4. Conclusion

Throughout this paper, sufficient conditions for the design of a state-dependent switching function assuring global asymptotic stability of a limit cycle $\mathcal{X}_e^*$, satisfying a given steady-state criterion, are provided for discrete-time switched affine systems. The proposed methodology consists in determining a subset of candidate limit cycles $\mathcal{X}_s$ from where an optimal one $\mathcal{X}_e^*$ is determined by minimising an $\mathcal{H}_2$ or $\mathcal{H}_\infty$ guaranteed cost. These conditions are based on a time-varying convex Lyapunov function being, to the best to the authors' knowledge, the first ones to deal with the problem under consideration. Differently from results dealing with practical stability where the steady-state behaviour is unpredictable, the limit cycle approach allows the designer to assure a suitable performance in the steady and transient states. A practical application example was used for control validation. An interesting topic for further investigation is the generalisation of these results to cope with sampled-data control and model uncertainties.

References

- Allerhand, L. I., & Shaked, U. (2013). Robust state-dependent switching of linear systems with dwell time. *IEEE Transactions on Automatic Control*, 58, 994–1001.
- Benmiloud, M., Benalia, A., Djemai, M., & Defoort, M. (2019). On the local stabilization of hybrid limit cycles in switched affine systems. *IEEE Transactions on Automatic Control*, 64, 841–846.
- Bittanti, S., & Colaneri, P. (2009). *Periodic systems: Filtering and control*. Springer-Verlag London.
- Cardim, R., Teixeira, M. C., Assunção, E., & Covacic, M. R. (2009). Variable-structure control design of switched systems with an application to a DC–DC power converter. *IEEE Transactions on Industrial Electronics*, 56, 3505–3513.
- Daiha, H. R., Egidio, L. N., Deaecto, G. S., & Geromel, J. C. (2017). $\mathcal{H}_\infty$ state feedback control design of discrete-time switched linear systems. In: *IEEE Conf. on Dec. and Control*. pp. 5882–5887.
- Dargahi, V., Sadigh, A. K., Pahlavani, M. R. A., & Shoulaie, A. (2012). DC (direct current) voltage source reduction in stacked multicell converter based energy systems. *Energy*, 46, 649–663.
- Deaecto, G. S. (2016). Dynamic output feedback $\mathcal{H}_\infty$ control of continuous-time switched affine systems. *Automatica*, 71, 44–49.
- Deaecto, G. S., Fioravanti, A. R., & Geromel, J. C. (2013). Suboptimal switching control consistency analysis for discrete-time switched linear systems. *European Journal of Control*, 19, 214–219.
- Deaecto, G. S., & Geromel, J. C. (2017). Stability analysis and control design of discrete-time switched affine systems. *IEEE Transactions on Automatic Control*, 62, 4058–4065.
- Deaecto, G. S., & Geromel, J. C. (2018). Stability and performance of discrete-time switched linear systems. *Systems & Control Letters*, 118, 1–7.
- Deaecto, G. S., Geromel, J. C., & Daafouz, J. (2011). Dynamic output feedback $\mathcal{H}_\infty$ control of switched linear systems. *Automatica*, 47, 1713–1720.
- Deaecto, G. S., & Santos, G. C. (2015). State feedback $\mathcal{H}_\infty$ control design of continuous-time switched affine systems. *IET Control Theory & Applications*, 9, 1511–1516.
- Egidio, L. N., Daiha, H. R., Deaecto, G. S., & Geromel, J. C. (2017). DC motor speed control via buck-boost converter through a state dependent limited frequency switching rule. In: *IEEE Conf. on Dec. and Control*. pp. 2072–2077.
- Egidio, L. N., & Deaecto, G. S. (2019). Novel practical stability conditions for discrete-time switched affine systems. *IEEE Transactions on Automatic Control*, 64(11), 4705–4710.
- Fiacchini, M., & Jungers, M. (2014). Necessary and sufficient condition for stabilizability of discrete-time linear switched systems: A set-theory approach. *Automatica*, 50, 75–83.
- Geromel, J. C., & Colaneri, P. (2006). Stability and stabilization of discrete time switched systems. *International Journal of Control*, 79, 719–728.
- Hauwoigne, P., Riedinger, P., & Jung, C. (2011). Switched affine systems using sampled-data controllers: Robust and guaranteed stabilization. *IEEE Transactions on Automatic Control*, 56, 2929–2935.
- Hetel, L., & Fridman, E. (2013). Robust sampled-data control of switched affine systems. *IEEE Transactions on Automatic Control*, 58, 2922–2928.
- Patino, D., Riedinger, P., & Ruiz, F. (2010). A predictive control approach for DC–DC power converters and cyclic switched systems. In: *IEEE Int. Conf. on Ind. Technol.* pp. 1259–1264.
- Sanchez, C. A., Garcia, G., Sabrina, H., Heemels, W. P. M. H., & Zaccarian, L. (2019). Practical stabilisation of switched affine systems with dwell-time guarantees. *IEEE Transactions on Automatic Control*, 64, 4811–4817.
- Seatzu, C., Corona, D., Giua, A., & Bemporad, A. (2006). Optimal control of continuous-time switched affine systems. *IEEE Transactions on Automatic Control*, 51, 726–741.
- Sferlazza, A., Albea-Sanchez, C., Martinez-Salamero, L., Garcia, G., & Alonso, C. (2020). Min-type control strategy of a DC–DC synchronous boost converter. *IEEE Transactions on Industrial Electronics*, 67, 3167–3179.
- Souza, M., Deaecto, G. S., Geromel, J. C., & Daafouz, J. (2014). Self-triggered linear quadratic networked control. *Optimal Control Applications & Methods*, 35, 524–538.
- Veer, S., & Poulakakis, I. (2019). Switched systems with multiple equilibria under disturbances: Boundedness and practical stability. *IEEE Transactions on Automatic Control*, <http://dx.doi.org/10.1109/TAC.2019.2931941>.



Lucas N. Egidio received the B.Sc. degree in Control and Automation Engineering in 2015, the M.Sc. degree in Mechanical Engineering in 2016, and the Ph.D. degree in Mechanical Engineering in 2020 from the University of Campinas (UNICAMP), Brazil. In 2018, he spent 6 months at the University of California, Santa Barbara, as a Visiting Scholar. Currently, he is a Postdoctoral Fellow at Linköping University, Sweden. His research interests include switched systems, convex optimisation and control applications in power electronics.



Helder R. Daiha was born in Tremembé, Brazil, in 1989. He received the B.Sc. degree in Mechanical Engineering from the University of Campinas (UNICAMP), Brazil, in 2015, and the M.Sc. degree in Mechanical Engineering from the same university in 2016. He is currently a Ph.D. candidate in Mechanical Engineering at UNICAMP. His research interests are convex optimisation, analysis and control design of switched systems, hybrid systems and robust control.



Grace S. Deaecto was born in São Paulo, SP, Brazil. She received the B.Sc. degree in Electrical Engineering from the São Paulo State University (UNESP), Ilha Solteira, Brazil, in 2005 and the M.Sc. and Ph.D. degrees in Electrical Engineering from the School of Electrical and Computer Engineering (FEEC), University of Campinas (UNICAMP), Brazil in 2007 and 2010, respectively. She joined the School of Mechanical Engineering at UNICAMP in 2013, where she is currently Assistant Professor. She is Class 1D Researcher of the Brazilian National Council for Scientific and Technological Development (CNPq) and member of the IFAC Technical Committee on Robust Control. She is serving as an Associate Editor for the IFAC Journal, Nonlinear Analysis: Hybrid Systems since 2018. She is co-author of the book *Análise Linear de Sinais: Teoria, ensaios práticos e exercícios* (in Portuguese), Edgard Blücher, Ltda, 2019. Her research interests include convex optimisation, robust control system and control design of switched systems.